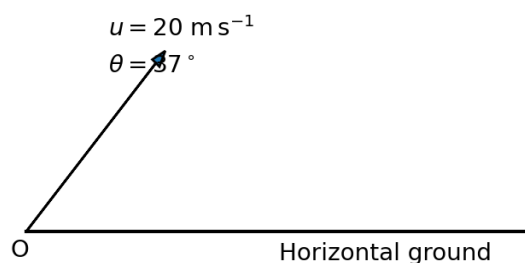


JEE Main Physics – Solutions

Detailed solutions corresponding to the 10 questions.

Solution 1



Given $u = 20 \text{ m s}^{-1}$, $\theta = 37^\circ$, $g = 10 \text{ m s}^{-2}$.

$u_x = u \cos\theta = 20 \times (4/5) = 16 \text{ m s}^{-1}$, $u_y = u \sin\theta = 20 \times (3/5) = 12 \text{ m s}^{-1}$.

(a) Time of flight:

$$T = 2u_y/g = 2 \times 12/10 = 2.4 \text{ s}.$$

(b) Range:

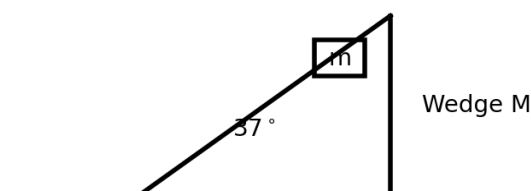
$$R = u_x T = 16 \times 2.4 = 38.4 \text{ m}.$$

(c) Maximum height:

$$H = u_y^2/(2g) = 12^2/(20) = 7.2 \text{ m}.$$

Answer: $T = 2.4 \text{ s}$, $R = 38.4 \text{ m}$, $H = 7.2 \text{ m}$.

Solution 2



For a smooth wedge–block system,
acceleration of wedge:

$$A = (m g \sin\theta \cos\theta) / (M + m \sin^2\theta).$$

Using $m = 2 \text{ kg}$, $M = 8 \text{ kg}$, $\sin 37^\circ = 3/5$, $\cos 37^\circ = 4/5$:

$$A = [2 \times 10 \times (3/5) \times (4/5)] / [8 + 2 \times (9/25)]$$

$$= 9.6 / 8.72 \approx 1.10 \text{ m s}^{-2}.$$

Acceleration of block relative to wedge:

$$\begin{aligned} a_{\text{rel}} &= [(M+m)g \sin\theta] / [M + m \sin^2\theta] \\ &= [10 \times 10 \times (3/5)] / 8.72 \\ &= 60 / 8.72 \approx 6.88 \text{ m s}^{-2} \text{ down the incline.} \end{aligned}$$

Horizontal acceleration of block w.r.t. ground:

$$\begin{aligned} a_x &= A + a_{\text{rel}} \cos\theta \\ &\approx 1.10 + 6.88 \times (4/5) \\ &\approx 6.60 \text{ m s}^{-2} \text{ to the right.} \end{aligned}$$

Answer:

- (a) 1.10 m s^{-2} ,
- (b) 6.88 m s^{-2} down the plane,
- (c) 6.60 m s^{-2} rightward.

Solution 3

Work done:

$$\begin{aligned} W &= \int_0^3 (4x^2 - 6x + 2) \, dx \\ &= [(4/3)x^3 - 3x^2 + 2x]_0^3 \\ &= 36 - 27 + 6 \\ &= 15 \text{ J.} \end{aligned}$$

Since the particle starts from rest, initial kinetic energy = 0.

Thus,

$$\begin{aligned} (1/2)mv^2 &= 15 \\ (1/2)(2)v^2 &= 15 \\ v^2 &= 15 \\ v &= \sqrt{15} \approx 3.87 \text{ m s}^{-1}. \end{aligned}$$

Answer: speed = $\sqrt{15} \approx 3.87 \text{ m s}^{-1}$.

Solution 4

At distance $r = 3R$ from Earth's centre:

$$g' = GM/(3R)^2 = g/9.$$

(a) Acceleration due to gravity at orbit = $g/9$.

For circular motion, $v = \sqrt{GM/r}$.

Since $GM = gR^2$,

$$v = \sqrt{gR^2/3R} = \sqrt{gR/3}.$$

(b) Orbital speed = $\sqrt{gR/3}$.

Time period:

$$T = 2\pi r / v = 2\pi(3R) / \sqrt{(gR/3)} \\ = 2\pi\sqrt{(27R/g)}.$$

(c) Time period = $2\pi\sqrt{(27R/g)}$.

Answer:

(a) $g/9$

(b) $\sqrt{(gR/3)}$

(c) $2\pi\sqrt{(27R/g)}$.

Solution 5

At constant pressure for 1 mole:

$$\Delta T = 600 - 300 = 300 \text{ K}.$$

(a) Work done:

$$W = nR\Delta T = 1 \times 8.314 \times 300 = 2494.2 \text{ J}.$$

(b) Change in internal energy:

$$\Delta U = (3/2)nR\Delta T = (3/2) \times 8.314 \times 300 = 3741.3 \text{ J}.$$

(c) Heat supplied:

$$Q = \Delta U + W = 3741.3 + 2494.2 = 6235.5 \text{ J}.$$

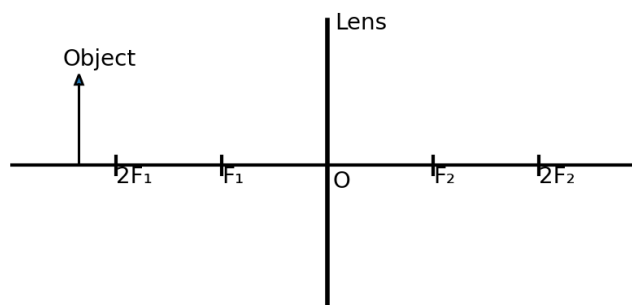
Answer:

$$W = 2.49 \times 10^3 \text{ J},$$

$$\Delta U = 3.74 \times 10^3 \text{ J},$$

$$Q = 6.24 \times 10^3 \text{ J}.$$

Solution 6



For a convex lens, $f = +20 \text{ cm}$, object distance $u = -30 \text{ cm}$.

Lens formula:

$$1/f = 1/v - 1/u$$

$$1/20 = 1/v + 1/30$$

$$1/v = 1/20 - 1/30 = (3-2)/60 = 1/60$$

$$v = +60 \text{ cm.}$$

(a) Image distance = 60 cm on the right side.

Magnification:

$$m = v/u = 60/(-30) = -2.$$

(b) Magnification = -2.

(c) Negative magnification means real and inverted image; $|m| = 2$ means enlarged 2 times.

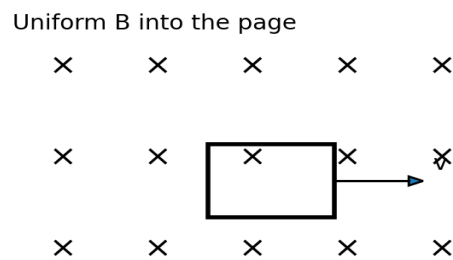
Answer:

(a) 60 cm

(b) -2

(c) real, inverted, magnified.

Solution 7



When the loop enters the field, changing flux induces emf:

$$\varepsilon = B \frac{d\Phi}{dt}$$

where $\frac{d\Phi}{dt}$ is the side perpendicular to velocity = 20 cm = 0.2 m.

Thus,

$$\varepsilon = 0.5 \times 0.2 \times 5 = 0.5 \text{ V.}$$

(a) Induced emf = 0.5 V.

Current:

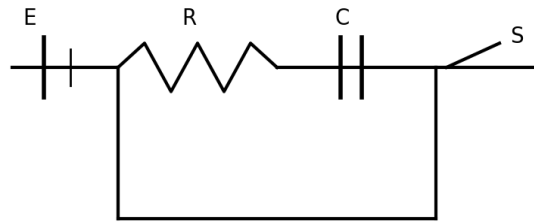
$$I = \varepsilon/R = 0.5 / 0.2 = 2.5 \text{ A.}$$

(b) Induced current = 2.5 A.

Answer:

$$\varepsilon = 0.5 \text{ V, } I = 2.5 \text{ A.}$$

Solution 8



Given $R = 3 \text{ k}\Omega = 3000 \text{ }\Omega$, $C = 200 \text{ }\mu\text{F} = 200 \times 10^{-6} \text{ F}$, $E = 12 \text{ V}$.

(a) Time constant:

$$\tau = RC = 3000 \times 200 \times 10^{-6} = 0.6 \text{ s}.$$

(b) Initial current:

$$I_0 = E/R = 12/3000 = 4 \times 10^{-3} \text{ A} = 4 \text{ mA}.$$

(c) Charge at $t = \tau$:

$$\begin{aligned} q(\tau) &= CE(1 - e^{-1}) \\ &= (200 \times 10^{-6})(12)(1 - 1/e) \\ &= 2.4 \times 10^{-3} \times 0.632 \\ &\approx 1.52 \times 10^{-3} \text{ C}. \end{aligned}$$

Answer:

$$\tau = 0.6 \text{ s},$$

$$I_0 = 4 \text{ mA},$$

$$q(\tau) \approx 1.52 \text{ mC}.$$

Solution 9

Photon energy:

$$E = hc/\lambda = 1240/400 = 3.1 \text{ eV}.$$

(a) Incident photon energy = 3.1 eV.

Maximum kinetic energy:

$$K_{\text{max}} = E - \phi = 3.1 - 2.2 = 0.9 \text{ eV}.$$

(b) $K_{\text{max}} = 0.9 \text{ eV}$.

Stopping potential:

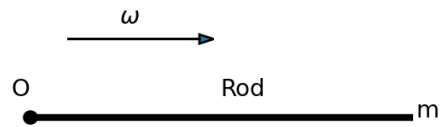
$$eV_0 = K_{\text{max}} \Rightarrow V_0 = 0.9 \text{ V}.$$

(c) Stopping potential = 0.9 V.

Answer:

3.1 eV, 0.9 eV, 0.9 V.

Solution 10



Conserve angular momentum about hinge O during the collision.

Initial angular momentum of particle:

$$L_i = m v L = 1 \times 6 \times 1.2 = 7.2 \text{ kg m}^2 \text{ s}^{-1}.$$

Total moment of inertia after sticking:

$$I_{\text{total}} = I_{\text{rod}} + mL^2$$

$$= (1/3)ML^2 + mL^2$$

$$= (1/3)(3)(1.2)^2 + 1 \times (1.2)^2$$

$$= 1.44 + 1.44$$

$$= 2.88 \text{ kg m}^2.$$

Therefore,

$$\omega = L_i / I_{\text{total}} = 7.2 / 2.88 = 2.5 \text{ rad s}^{-1}.$$

Answer: angular speed = 2.5 rad s^{-1} .